

Project Documentation

End to End customer churn Analytics and Prediction System

1. Project Overview

Project Name: End to End customer churn Analytics and Prediction System

Description:

This project is an end-to-end data solution that collects, processes, analyzes, and visualizes data to generate meaningful insights and predictions. It also provides real-time predictions using a machine learning model through a Flask API.

2. Objective

- To analyze raw data and extract useful insights
 - To build predictive models (classification & regression)
 - To visualize data using dashboards
 - To deploy a model using Flask API
 - To simulate real-world business intelligence workflow
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3. Tech Stack

Programming & Analysis

- Python
- Pandas, NumPy

Visualization

- Matplotlib
- Seaborn

Database

- MySQL

Machine Learning

- Scikit-learn

NLP

- NLTK / Scikit-learn

Dashboard

- Power BI / Tableau

Backend

- Flask
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4. Project Workflow

Step 1: Data Collection

- Dataset collected from CSV / Database
 - Example: Customer, Sales, or E-commerce data
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Step 2: Data Preprocessing

- Handle missing values
 - Remove duplicates
 - Convert data types
 - Handle outliers
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Step 3: Exploratory Data Analysis (EDA)

- Understand data distribution
- GroupBy and aggregation
- Identify patterns and trends

Step 4: Data Visualization

- Bar charts (comparison)
- Line charts (trend analysis)
- Pie charts (distribution)
- Heatmap (correlation analysis)

Step 5: SQL Integration

- Store data in MySQL •
- Perform queries: ○
SELECT ○ WHERE ○
JOIN ○ GROUP BY ○
ORDER BY

Step 6: Machine Learning

Classification:

- Logistic Regression
- Random Forest

Regression:

- Linear Regression

Steps:

- Feature selection
- Train-test split
- Model training
- Model evaluation

Evaluation Metrics:

- Accuracy
- Confusion Matrix
- MAE / RMSE

Step 7: NLP (Optional)

- Text preprocessing • Tokenization
- Stop word removal
- TF-IDF / Count Vectorizer
- Sentiment Analysis

Step 8: Dashboard Creation

- KPIs (Sales, Profit, Customers)
- Filters and slicers
- Interactive charts • Business insights

Step 9: Flask API

- Create API endpoint
- Accept input in JSON format
- Return prediction

Example:

Input:

```
{  
  "input": [45, 50000, 2] }
```

Output:

```
{  
  "prediction": 1  
}
```

Step 10: Deployment (Optional)

- Deploy Flask app on cloud
 - Make API publicly accessible
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5. Key Features

- Data cleaning and transformation
 - Business insights generation
 - Predictive modeling
 - Interactive dashboard
 - Real-time prediction API
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6. Expected Outcomes

- Identify trends and patterns
 - Understand customer behavior
 - Predict future outcomes
 - Provide business recommendations
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7. Use Cases

- Customer churn prediction
 - Sales forecasting
 - Product performance analysis
 - Customer sentiment analysis
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8. Folder Structure

```
project/ |  
├── data/  
├── notebooks/  
├── sql/  
├── models/  
├── app.py  
├── requirements.txt  
└── dashboard/
```

9. Conclusion

This project demonstrates a complete data science lifecycle, from raw data processing to deployment. It helps in understanding how real-world business intelligence systems work and prepares for roles like Data Analyst, Data Scientist, and ML Engineer.

Final Summary

This project converts raw data into actionable insights and predictions using data analysis, machine learning, and business intelligence tools, making it a complete industry-level solution.
